

GLASGOW, KY, USA

WMO: 722322

Lat: 37.033N

Lon: 85.950W

Elev: 218

StdP: 98.73

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 00361

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.6	-8.7	-19.8	0.7	-7.9	-17.0	0.9	-7.2	9.4	9.5	8.5	9.6	1.4	350	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	34.0	24.0	32.9	23.5	32.1	23.4	25.8	30.8	25.2	30.4	24.6	29.4	2.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.8	20.4	27.5	23.9	19.2	26.9	23.0	18.2	26.3	81.3	30.9	78.0	30.3	75.4	29.5	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.9	6.9	5.6	-16.1	36.1	5.1	2.3	-19.8	37.8	-22.8	39.1	-25.7	40.5	-29.4	42.2		
			-16.9	26.8	4.7	1.3	-20.2	27.8	-22.9	28.5	-25.6	29.2	-29.0	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	2.3	4.0	9.1	14.5	19.7	24.0	25.6	24.9	21.7	15.0	8.2	4.5
	DBStd	9.62	6.52	6.36	6.21	5.20	4.22	2.64	2.43	2.53	3.58	5.17	5.33	5.77
	HDD10.0	844	250	183	94	18	1	0	0	0	0	13	95	189
	HDD18.3	2239	496	401	291	135	39	1	0	0	12	131	306	428
	CDD10.0	2489	12	17	66	152	299	420	482	462	351	167	42	19
	CDD18.3	841	0	0	4	18	79	171	224	203	113	26	2	0
	CDH23.3	7828	0	1	31	169	689	1604	2126	1851	1083	266	9	0
	CDH26.7	3141	0	0	2	23	202	648	943	800	449	73	1	0
	WSAvg	2.0	2.6	2.7	2.7	2.5	1.7	1.5	1.4	1.2	1.2	1.7	2.0	2.3
Precipitation	PrecAvg	1294	111	96	126	110	136	113	118	82	95	91	91	122
	PrecMax	1547	201	174	351	182	315	233	216	201	206	188	148	282
	PrecMin	971	54	31	66	46	46	42	22	16	21	10	21	56
	PrecStd	168	33	36	66	40	66	47	42	43	57	50	32	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	21.4	27.0	28.9	32.3	34.1	36.1	36.2	34.9	31.3	24.3	20.9
		MCWB	13.2	15.3	17.5	19.0	22.1	23.5	23.4	26.4	22.6	23.1	16.6	15.8
	2%	DB	17.4	18.9	23.9	27.2	30.2	32.8	33.8	33.9	32.9	28.6	22.1	18.7
		MCWB	12.8	14.1	16.2	18.1	21.1	22.8	24.5	24.5	22.6	20.5	15.4	15.4
	5%	DB	15.7	17.0	21.9	25.3	28.9	32.1	32.6	32.5	31.4	26.4	19.6	16.3
		MCWB	11.6	12.5	15.1	16.7	20.4	22.7	24.2	23.9	22.4	18.6	14.4	13.1
	10%	DB	12.9	14.2	19.0	23.4	27.5	30.6	31.6	31.2	29.2	23.9	17.4	14.0
		MCWB	9.0	10.3	13.3	16.1	20.0	22.5	23.8	23.5	21.4	17.6	12.9	11.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.7	17.2	18.2	20.6	23.4	25.7	26.9	27.7	25.7	24.4	18.6	17.9
		MCDB	17.5	19.3	23.9	26.6	28.9	31.2	29.9	33.5	30.8	27.8	21.2	19.6
	2%	WB	13.8	15.3	17.3	19.5	22.3	24.9	25.7	26.2	24.0	22.1	17.2	16.0
		MCDB	16.5	18.5	22.5	24.7	27.9	30.4	30.3	31.6	29.1	26.3	20.6	18.0
	5%	WB	12.2	13.5	16.0	18.4	21.6	24.0	25.2	25.2	23.1	20.3	15.3	14.0
		MCDB	15.1	15.8	20.5	23.2	27.0	29.3	30.5	30.0	28.7	24.0	18.0	16.2
	10%	WB	9.7	11.1	14.6	17.3	20.7	23.2	24.7	24.4	22.3	18.6	13.5	11.7
		MCDB	13.4	14.1	18.4	21.7	25.8	28.3	29.6	28.6	27.6	23.1	16.7	14.1
Mean Daily Temperature Range	5% DB	MDBR	9.7	10.2	11.5	12.6	11.9	11.6	10.9	11.4	12.7	12.8	11.8	9.5
		MCDBR	11.2	12.5	13.6	14.3	13.4	13.1	12.3	13.1	15.0	14.9	13.9	11.4
	5% WB	MCWBR	8.9	9.4	8.1	7.4	5.6	5.2	4.5	5.4	5.7	7.2	8.7	9.0
		MCWBR	10.5	10.9	11.8	11.5	11.3	11.1	10.6	11.2	12.7	12.7	10.9	10.2
Clear-Sky Solar Irradiance	taub		0.313	0.322	0.360	0.402	0.446	0.467	0.499	0.508	0.421	0.366	0.331	0.308
		taud	2.470	2.431	2.371	2.251	2.174	2.149	2.079	2.040	2.265	2.409	2.466	2.492
	Ebn at Noon		877	913	903	879	842	822	791	774	829	850	848	859
		Edh at Noon	83	97	113	135	148	152	162	165	125	98	83	76
All-Sky Solar Radiation	RadAvg		2.06	2.69	3.78	5.01	5.59	6.22	5.91	5.50	4.60	3.52	2.48	1.78
	RadStd		0.21	0.36	0.36	0.37	0.51	0.36	0.34	0.29	0.42	0.41	0.26	0.17

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+130	+84

Nomenclature: See separate page